

1. An hypothetical metal with an ultimate tensile strength of 450 MPa and a 0.2% offset yield strength of 250MPa is loaded in tension in a mechanical testing machine. A force of 3.5 kN is applied along the long axis of the sample and it is found to elongate by 1.2 mm while this load is applied. The load is then increased to 10 kN and the total elongation is measured to be 6.0 mm. While the 10 kN load is applied the stress and strain are known to be 400 MPa and 0.5%, respectively.

(a) (6 points) Determine the Young's modulus of this hypothetical metal.

- (b) (3 points) In point form, identify how the mechanical properties of this alloy would be expected to have changed following the application of the 10 kN load.

2. (5 points) the density of silver at 800 °C is 9.5 g cm^{-3} and the energy for vacancy formation at this temperature is 1.1 eV/atom. Calculate the equilibrium number of vacancies per cubic metre for silver at this temperature.

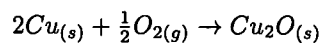
3. (5 points) Determine the enthalpy change for the combustion of 1 mole of gaseous methane to form gaseous carbon dioxide and liquid water.

4. Silicon is doped with phosphorus to create an extrinsic semiconductor. The electron mobility in silicon is $0.14 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ and the hole mobility is $0.05 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$.

(a) (4 points) Determine the electrical conductivity of this semiconductor if the dopant concentration is 10^{21} m^{-3} .

(b) (4 points) Another silicon based semiconductor has a room temperature conductivity of $4 \times 10^{-4} \Omega^{-1} \text{ m}^{-1}$ and a hole concentration is $1.314 \times 10^{16} \text{ m}^{-3}$. Determine if this semiconductor is intrinsic, p-type, or n-type.

5. Consider the reaction of solid copper with gaseous oxygen to yield solid cuprous oxide, as shown below.



- (a) (3 points) Determine the standard Gibbs energy change for this reaction.

- (b) (5 points) Determine the enthalpy change for the melting of one mole of cuprous oxide.

- (c) (4 points) If the entropy change associated with the melting of cuprous oxide is $9.16 \text{ J mol}^{-1} \text{ K}^{-1}$, determine the melting temperature of cuprous oxide in degrees Celsius.

6. (8 points) The Wikipedia page for silver bromide lists its mass density as 6.473 g cm^{-3} . and indicates that it forms the rock salt crystal structure. Another Wikipedia page lists the ionic radii for silver and bromine as 129 pm and 182 pm, respectively. Using these radii, determine the theoretical density for silver bromide and conclude as to whether Wikipedia's data is consistent, based on your calculation.

7. (5 points) A bridge is (unwisely) constructed from unreinforced concrete. The bridge covers a 2.5 m span and has a rectangular cross-section with height of 30 cm and width of 1.3 m. If the bridge is intended as a pedestrian bridge and we assume a mass of 100 kg per person, how many people could stand in the middle of this bridge to take a selfie without it fracturing? Assume an acceleration due to gravity of 9.81 N kg^{-1}

8. (10 points) Taking a lattice parameter for crystalline silicon of 5.4×10^{-10} m, determine the theoretical density of crystalline silicon.

9. (10 points) Show that the size of the cation that may occupy the octahedral interstitial site ranges from a minimum of 41% to a maximum of 73% the size of the anions.

Part B: Multiple Choice (1 points per question)

1. An increase in the Gibbs energy corresponds to which of the following?
 - (a) A decrease in the entropy of the universe.
 - (b) An increase in the entropy of the universe.

2. Which of the following can be thought of as an accurate description of the structure of crystalline silicon?
 - (a) An FCC lattice of Si atoms with Si atoms occupying all of the octahedral interstitial sites.
 - (b) An FCC lattice of Si atoms with Si atoms occupying half of the tetrahedral interstitial sites.
 - (c) An FCC lattice of Si atoms with Si atoms occupying half of the octahedral interstitial sites.
 - (d) An FCC lattice of Si atoms with Si atoms occupying all of the tetrahedral interstitial sites.

3. An hypothetical metal alloy supports a stress of 783 MPa and experiences a total strain of 0.02 while this load is applied. When the load is removed a permanent strain of 0.01 is observed. What is the Young's modulus of this metal?
 - (a) 90.05 GPa
 - (b) 78.3 GPa
 - (c) 17.11 GPa
 - (d) 39.15 GPa

4. Which of the following statements regarding spontaneity is true?
 - (a) A process involving a decrease in the system entropy can never be spontaneous.
 - (b) The sole requirement for the spontaneity of a process is the increase in entropy of the universe.
 - (c) An exothermic process is always spontaneous.
 - (d) A spontaneous process will necessarily proceed rapidly.

5. How is it that carbon can form 4 identical bonds with different hydrogen atoms in a molecule of methane when the carbon atom has an electron configuration of $[He]2s^2 2p^2$?
- (a) There are actually two different bond strengths, from the s and p electrons
 - (b) None of these are correct.
 - (c) The single s shell is combined with the three p shells to create four hybridized orbitals
 - (d) The two s electrons are combined with two of the p shells to create four hybridized orbitals
6. Which of the following statements is false?
- (a) The electron mobility is generally greater than the hole mobility.
 - (b) In n-type semiconductors, an atom with more valence electrons than the host atom is added to the semiconductor, thereby creating holes as the majority charge carriers.
 - (c) The donor level in an n-type semiconductor is expected to be within the band gap, close to the conduction band.
 - (d) Doping silicon with gallium results in an extrinsic p-type semiconductor.
7. Sugar is dissolved in water at room temperature to create a syrup having a composition of about 10 wt% sugar. This system is then cooled until solid water begins to form. Which of the following is true of the syrup that remains in equilibrium with the solid ice?
- (a) There will be no syrup present under these conditions.
 - (b) It's composition remains at 10 wt%.
 - (c) It has a composition greater than 10 wt%.
 - (d) It has a composition less than 10 wt%.
8. Which of the following is not a requirement for optical transparency?
- (a) An amorphous microstructure.
 - (b) An absence of internal scattering events.
 - (c) A high band gap.

9. Which of the following thermodynamic quantities are state variables: P, q, S, T, U, V, W?
- (a) All of these except q and W.
 - (b) All of these.
 - (c) None of these.
 - (d) Only q and W.
10. Sugar is dissolved in water at room temperature to create a single phase liquid syrup. This system is cooled until the entire system is solid. Which of the following is true of this system?
- (a) This system will never become entirely solid.
 - (b) It consists of essentially pure water and pure sugar, both solid.
 - (c) None of these options are true.
 - (d) It consists of a single solid phase of sugar dissolved in solid water.
11. Which of the following statements is true?
- (a) The yield strength of an alloy is generally lower than that of either of the constituent elements.
 - (b) The coordination number for cations in the rock salt crystal structure is 8
 - (c) The number of vacancies in a metal decreases with increasing temperature.
 - (d) The yield strength of a metal decreases with increasing grain size.
12. In the First Law of Thermodynamics, the sign convention is such that work performed on the system carries which sign?
- (a) Positive.
 - (b) Negative.
 - (c) Work done on the system does not change the internal energy of the system.

13. The solidification of liquid water to solid water ice is an example of which of the following types of reaction?
- (a) Endothermic, positive entropy change
 - (b) Endothermic, negative entropy change
 - (c) Exothermic, negative entropy change
 - (d) Exothermic, positive entropy change
14. Which of the following statements is true?
- (a) The absolute entropy of the solid phase of a substance will always be higher than that of the gas phase.
 - (b) The Gibbs energy change for the mixing of two gases is always positive.
 - (c) The molar entropy of fusion is always positive.
 - (d) The enthalpy of combustion has a positive value.
15. Amorphous, or glassy, metals are an exciting area of current research. The band gap of these materials would be expected to be which of the following?
- (a) No band gap.
 - (b) A large band gap.
 - (c) A small band gap.
 - (d) Unable to determine.
16. The dissolution of sugar into liquid water must be which of the following types of reactions?
- (a) Exothermic, positive entropy change
 - (b) Endothermic, negative entropy change
 - (c) Exothermic, negative entropy change
 - (d) Endothermic, positive entropy change

17. Which of the following statements about the Second Law of Thermodynamics is *false*?

- (a) In a spontaneous process, the entropy of the system must increase.
- (b) In a spontaneous process, the entropy of the universe must increase.
- (c) The entropy of the universe is equal to the sum of the entropy of the system and the entropy of the surroundings.
- (d) In a spontaneous process, the entropy of the surroundings may increase or decrease.

18. Which of the following statements about thermodynamics is *false*?

- (a) Elements in their most stable state under standard conditions have zero entropy.
- (b) Absolute entropy increases with increases in temperature.
- (c) The absolute entropy of a pure substance at standard conditions can be evaluated.
- (d) The entropy of all pure substances approaches zero as the temperature approaches 0 K.

19. Which of the following bond types is considered non-directional?

- (a) Ionic
- (b) Covalent
- (c) Metallic

20. Sugar is dissolved in water at 80 °C until no additional sugar can dissolve. The system is then allowed to cool to room temperature and reach equilibrium. What phase(s) do you expect to be present?

- (a) Liquid water plus solid sugar.
- (b) Liquid water plus liquid sugar.
- (c) Liquid syrup.
- (d) Liquid syrup plus solid sugar

You may remove these pages from your test.

Constants

$$1 \text{ atm} = 101.325 \text{ kPa} = 1.01325 \text{ bar} = 14.696 \text{ psi} \quad N_A = 6.022 \times 10^{23} \text{ mol}^{-1} \quad e = 1.602 \times 10^{-19} \text{ C}$$

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J} \quad \epsilon_0 = 8.854 \times 10^{-12} \frac{\text{F}}{\text{m}} \quad R = 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} = 0.082067 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}$$

$$T(\text{K}) = T(^{\circ}\text{C}) + 273.15 \quad k = 8.62 \times 10^{-5} \frac{\text{eV}}{\text{atom} \cdot \text{K}} \quad k = 1.38 \times 10^{-23} \frac{\text{J}}{\text{atom} \cdot \text{K}} \quad F = 96486 \text{ C} \cdot \text{mol}^{-1}$$

Microstructure

$$\text{LD} = \#/\text{Length} \quad \text{LPF} = \frac{\text{length of atoms}}{\text{length of vector}} \quad \text{PD} = \#/\text{Area} \quad \text{PPF} = \frac{\text{area of atoms}}{\text{area of plane}} \quad V = \frac{4}{3} \pi r^3$$

$$A = \pi r^2 \quad A_{\Delta} = \frac{1}{2} bh \quad \rho = \frac{n \cdot A}{V_C \cdot N_A} \quad \rho = \frac{m}{V} \quad \rho = \frac{n_A A_A + n_C A_C}{V_C N_A} \quad \text{APF} = \frac{V_s}{V_C}$$

$$N = \frac{N_A \rho}{A} \quad N_v = N \exp\left(-\frac{Q_v}{kT}\right) \quad a = 2\sqrt{2}R \quad a = \frac{4}{\sqrt{3}}R$$

Mechanical Behaviour

$$\sigma = \frac{F}{A} \quad \epsilon = \frac{\Delta l}{l_0} \quad \sigma = E \epsilon \quad \sigma_{3\text{-point bend}} = \frac{3FL}{2w \cdot h^2} x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Electrical Behaviour

$$\sigma = n|e|\mu_e + h|e|\mu_h \quad \sigma = n|e|\mu_e \quad \sigma = h|e|\mu_h$$

Thermodynamics

$$PV = nRT \quad \Delta U = q + w \quad \Delta U = q - P_{\text{ext}} \Delta V \quad H \equiv U + PV \quad G \equiv H - TS \quad \Delta S = \frac{q_{\text{rev}}}{T}$$

$$\text{For process at constant T: } \Delta G = \Delta H - T \Delta S \quad q = mc \Delta T = nC_p \Delta T \quad \Delta G = \Delta H - T \Delta S$$

$$W_{\text{phase}} = \frac{\text{length of opposite side of lever}}{\text{total length of lever}} \quad \text{For } aA + bB \rightarrow cC + dD, \quad Q = \frac{a_C^c \cdot a_D^d}{a_A^a \cdot a_B^b}$$

v_i : stoichiometric coefficient Assuming no phase change, constant C_p :

$$\Delta H_{\text{reaction}}^{\circ} = (\sum v_i \Delta H_{f,i}^{\circ})_{\text{products}} - (\sum v_i \Delta H_{f,i}^{\circ})_{\text{reactants}}$$

$$\Delta S_{\text{reaction}}^{\circ} = (\sum v_i \Delta S_{m,i}^{\circ})_{\text{products}} - (\sum v_i \Delta S_{m,i}^{\circ})_{\text{reactants}}$$

$$\Delta G_{\text{reaction}}^{\circ} = \Delta H_{\text{reaction}}^{\circ} - T \Delta S_{\text{reaction}}^{\circ}$$

$$\Delta G_{\text{reaction}} = \Delta G^{\circ} + RT \ln Q$$

$$\Delta G^{\circ} = -RT \ln K$$

Electrochemistry

$$E = E^{\circ} - \frac{RT}{nF} \ln Q$$

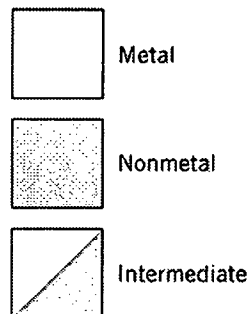
$$E = E^{\circ} - \frac{0.0592}{n} \ln Q, \text{ at } 25^{\circ}\text{C}$$

$$I = \frac{nc}{t} \quad w = nFE^{\circ}$$

																			Metal																																		
																			Nonmetal																																		
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																		Key			Atomic number			Symbol			Atomic weight																										
																		29			Cu			63.54																													

Key

29	Atomic number
Cu	Symbol
63.54	Atomic weight



Rare earth series

Actinide series

57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.35	63 Eu 151.96	64 Gd 157.25	65 Tb 158.92	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
89 Ac (227)	90 Th 232.04	91 Pa (231)	92 U 238.03	93 Np (237)	94 Pu (242)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (249)	99 Es (254)	100 Fm (253)	101 Md (256)	102 No (254)	103 Lw (257)

**STANDARD FORMATION ENTHALPY, STANDARD ENTROPY AND STANDARD
FORMATION GIBBS ENERGY AT 298.15 K**

Species	$\Delta_f H^\circ$ [kJ/mol]	S° [J/mol·K]	$\Delta_f G^\circ$ [kJ/mol]
C (s, graphite)	0	5.74	0
CH ₄ (g)	-74.81	186.2	-50.75
C ₂ H ₂ (g)	-83.9	200.93	
C ₃ H ₈ (g)	-103.8	269.9	-23.49
CaC ₂ (s)	-59.8	70.3	
CaF ₂ (s)	-1225	68.87	-1162
CaF ₂ (l)	-1186	92.6	
Ca(OH) ₂ (s)	-987.0	83.0	
CO ₂ (g)	-393.5	213.6	-394.4
Cu ₂ O (s)	-168.6	93.1	
Cu ₂ O (l)	-154.79		
Cu (s)		33.2	
Fe (s)	0	27.3	0
Fe ₂ O ₃ (s)	-824.2	87.4	
H ₂ (g)		130.68	
H ₂ O (g)	-241.8	188.7	-228.6
H ₂ O (l)	-285.8	69	
O ₂ (g)	0	205.0	0

MISCELLANEOUS ENTHALPIES

Substance	Reaction	ΔH [kJ/mol]
F-F	Bond dissociation	157
F	Electron affinity	-328
	F (g) → F ⁻ (g)	
Ca	Second Ionization energy	1734
	Ca (g) → Ca ²⁺ (g)	

SPECIFIC HEATS AND HEAT CAPACITIES

Substance	Specific Heat c [J/g·K]	Molar Heat Capacity C_p [J/mol·K]
CO ₂ (g)	0.843	37.1
Air (g)	1.0	
H ₂ O (g)	2.03	36.4
H ₂ O (l)	4.184	75.3
H ₂ O (s)	2.09	37.7

TEMPERATURES AND ENTHALPIES OF PHASE CHANGES

Substance	Melting Point [°C]	$\Delta_{fus} H^\circ$ [kJ/mol]	Boiling Point [°C]	$\Delta_{vap} H^\circ$ [kJ/mol]
Al	658	10.6	2467	284
Ca	851	9.33	1487	162
CH ₄	-182	0.92	-164	8.18
H ₂ O	0	6.01	100	40.7
Fe	1530	14.9	2735	354

